

Adaptive Speculative Decoding: An RL Approach to EAGLE3 Drafter Selection

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1 Revised Project Direction

Our project studies adaptive drafter selection for speculative decoding. In speculative decoding, a small drafter proposes future tokens and a larger target model verifies them in parallel. The speedup depends on drafter-target alignment: if the proposed tokens are likely under the target model, more tokens are accepted per verification step.

The original proposal focused on reducing the number of active drafters with an active-set penalty. After further analysis, we found that active-set size alone is not a reliable proxy for actual serving performance. If only one drafter is executed at a time, the direct latency cost is determined more by the selected drafter and serving implementation than by the number of candidate drafters. Therefore, we revise the project goal: rather than optimizing the number of active drafters, we will keep a fixed pool of domain-specialized drafters and compare RL-style selection policies for choosing which drafter to use.

This direction is motivated by two recent adaptive speculative decoding systems. MetaSD [1] shows that specialized drafters can have strong domain-dependent performance and formulates drafter selection as a multi-armed bandit problem. ATLAS [2] similarly argues that static speculators can lose alignment as workloads shift, although it addresses the issue through a static/adaptive speculator pair with runtime learning. Instead of assuming the same checkpoints are available, we plan to construct our own controlled drafter pool using EAGLE3 [4] and evaluate whether domain specialization appears in our setup.

2 Planned Setup

We will use one fixed target verifier model X , such as a Llama-3.1-class 7B–8B instruction model compatible with SGLang/SpecForge. For this target, we will train multiple EAGLE3 drafters from the same base architecture but on different domain-specific datasets:

- Y_{gsm8k} : trained on GSM8K-style math reasoning data.
- Y_{mmlu} : trained on MMLU-style broad knowledge and exam QA data.
- Y_{sharegpt} : trained on ShareGPT-style general instruction/chat data.
- Y_{all} : trained on the union of GSM8K, MMLU, and ShareGPT data.

To make the comparison fair, all four drafters will use comparable training budgets. Ideally, we will match the number of optimizer steps or the approximate number of training tokens. Since domain-specific datasets are smaller than the all-domain union, a domain-specific drafter may revisit its 500-example dataset for more epochs, while Y_{all} trains for fewer epochs over the 1500-example union.

3 Progress Since Proposal

The main progress since the proposal is a more concrete and implementable experimental design. We reviewed the MetaSD and EAGLE/EAGLE3 setting and concluded that the project should first build a controlled heterogeneous drafter pool, then evaluate selection policies over that pool.

We have fixed the planned evaluation structure. Each domain will have a small training subset and a held-out evaluation subset. The held-out split is important because evaluating on the same examples used for drafter fine-tuning would confound specialization with memorization. The milestone-stage target sizes are:

- 500 training examples per domain.
- 100–200 held-out evaluation examples per domain.
- 1500 total training examples for the all-domain drafter.

We have also completed the initial training and evaluation environment. EAGLE3 is a white-box speculative decoding method, so the drafter is trained using features from the fixed target model. Our current pipeline prepares domain data, trains an EAGLE3 drafter through SpecForge, serves the target-drafter pair with SGLang, and records average accept length under fixed decoding settings.

To validate the pipeline end to end, we first ran a 5-step SpecForge EAGLE3 smoke test on a local NVIDIA RTX 4060 GPU using Qwen3-0.6B and verified finite losses and checkpoint writing. We then ran a larger small-scale GSM8K pilot with 900 training examples and 100 held-out evaluation examples. With the number of draft tokens fixed to 4, average accept length improved from 1.0086 at step 1 to 1.2195 after 900 training steps. The output throughput in the same run also increased from 69.35 tokens/sec to 90.41 tokens/sec. This pilot does not yet establish cross-domain specialization, but it verifies the key milestone claim: the local training environment is working, EAGLE3 training produces a usable drafter checkpoint, and the acceptance-based serving metric improves after training.

The local NVIDIA RTX 4060 setup is sufficient for debugging and small-scale pilots, but it is not sufficient for the final experiment scale. We attempted a larger target and hit GPU memory limits during training. For the next stage, we plan to use an A100/H100 (or equivalent) GPU and scale up along three axes: a larger target model, larger domain datasets, and longer training schedules. This should allow the final domain-specialization matrix to be run under a more realistic target-model setting.

4 Evaluation Plan

The first evaluation artifact will be a domain-by-drafter matrix. Each row is an evaluation domain and each column is a trained EAGLE3 drafter. The primary metric is average accept length, with the number of draft tokens fixed to 4 for all comparisons:

$$\text{AcceptLength} = \frac{\text{number of generated output tokens}}{\text{number of target verification steps}}. \quad (1)$$

This measures how many tokens the system advances per target-model verification call, which is the quantity most directly tied to speculative decoding speedup. We will also record token-level acceptance rate, relative speedup over autoregressive decoding, and latency statistics if the serving stack exposes stable timing measurements.

The expected result is that a domain-specialized drafter can outperform Y_{all} on at least some matching domains, while Y_{all} may be more robust on average. If this pattern appears, it creates room for adaptive selection. If it does not appear, that negative result is still informative: it would suggest that all-domain EAGLE3 training is sufficient in our setting.

Eval domain	Y_{gsm8k}	Y_{mmlu}	Y_{sharegpt}	Y_{all}
GSM8K	TBD	TBD	TBD	TBD
MMLU	TBD	TBD	TBD	TBD
ShareGPT	TBD	TBD	TBD	TBD

Table 1: Planned domain-by-drafter matrix. Each entry will report average accept length with 4 draft tokens.

5 RL Selection Plan

After training and evaluating the four drafters, we will use the fixed drafter pool for online selection. The selection granularity is still under investigation. Per-request selection is easier to implement because the router chooses one drafter for the whole response. Per-step selection is closer to MetaSD because the policy can switch drafters at each speculative decoding round, but it may require deeper changes to the serving loop. We will decide between these options after checking the runtime integration cost.

The candidate RL and bandit methods are also tentative. We currently plan to start with simple baselines and add more methods only if the implementation remains stable:

- Uniform random selection.
- Greedy selection from observed rewards.
- UCB.
- Thompson sampling or EXP3 as optional alternatives.

The reward will initially be average accept length or an equivalent acceptance-based signal. If per-step selection is feasible, we may also consider denser alignment feedback similar to MetaSD’s block divergence reward.

6 Next Steps

The remaining work is:

1. Finalize the target model X .
2. Rent or otherwise secure an H100-class GPU for the full-scale training runs.
3. Build the domain-specific datasets and held-out evaluation splits.
4. Extend the validated SpecForge/EAGLE3 setup from the Qwen3-0.6B GSM8K pilot to larger models, larger datasets, and longer training schedules.
5. Train Y_{gsm8k} , Y_{mmlu} , Y_{sharegpt} , and Y_{all} with matched training budgets on the H100 setup.
6. Produce the domain-by-drafter matrix.
7. Decide whether dynamic selection will be implemented per request or per speculative decoding step.
8. Implement and compare a small set of RL/bandit selection policies.

References

- [1] T. Kim, H. Jung, and S.-Y. Yun. Multi-Drafter Speculative Decoding with Alignment Feedback. arXiv:2604.05417, 2026.

- [2] J. Wang et al. AdapTive-LeArning Speculator System (ATLAS): A New Paradigm in LLM Inference via Runtime-Learning Accelerators. Together AI Blog, 2025. <https://www.together.ai/blog/adaptive-learning-speculator-system-atlas>.
- [3] Y. Leviathan, M. Kalman, and Y. Matias. Fast Inference from Transformers via Speculative Decoding. ICML, 2023.
- [4] Y. Li et al. EAGLE-3: Scaling up Inference Acceleration of Large Language Models via Training-Time Test. arXiv:2503.01840, 2025.
- [5] D. Hendrycks et al. Measuring Massive Multitask Language Understanding. ICLR, 2021.
- [6] K. Cobbe et al. Training Verifiers to Solve Math Word Problems. arXiv:2110.14168, 2021.